

Smart Waste Collection and Monitoring System

Introduction

The Smart Waste Collection and Monitoring System's functional and non-functional requirements are outlined in this Software Requirements Specification document. Through automated scheduling, monitoring, complaint handling, and reporting, the system seeks to increase the effectiveness of waste management.

Project Charter

By substituting an automated digital platform for manual scheduling and reporting procedures, the Smart Waste Collection and Monitoring System aims to revolutionize waste management operations.

Requirements Elicitation

Informal interviews and surveys were carried out by the group with:

Barangay officials

Locals

Employees who collect waste

Synopsis of Results

Important Findings

Schedules for collection are frequently unknown to residents.

The documentation of complaint management is inadequate.

Reports on waste collection are processed by hand.

Centralized monitoring is necessary for administrators.

System Boundary

The proposed system's boundaries and scope are outlined in the Smart Waste Collection and Monitoring System. All digital procedures pertaining to waste collection scheduling, complaint handling, monitoring, and reporting within the barrio are covered by the system.

The system communicates with outside parties like residents, garbage collection workers, and administrators. Through a web-based interface, these users input data and receive outputs from the system.

Processes included within the system boundary are:

- User account management
- Waste collection scheduling
- Complaint submission and tracking
- Notification management
- Collection monitoring
- Report generation and analytics

Processes excluded from the system boundary are:

- Physical waste collection operations
- GPS tracking integration
- IoT sensor monitoring
- External payment processing systems
- Third-party logistics systems

Functional Requirements

ID	Requirement	Priority
FR-01	The system shall allow administrators to manage user accounts.	Must
FR-02	The system shall allow administrators to create collection schedules.	Must
FR-03	The system shall generate automated reports.	Must
FR-04	The system shall display analytics dashboards.	Should
FR-05	The system shall manage complaints and feedback.	Must

Non-Functional Requirements

ID	Requirement
NFR-01	The system shall load pages within 3 seconds.
NFR-02	The system shall use secure login authentication.
NFR-03	The system shall support mobile responsiveness.
NFR-04	The system shall maintain data backup procedures.
NFR-05	The system shall provide a user-friendly interface.

Stakeholder Analysis

Stakeholder	Role	Interest	Influence
Barangay Administrator	System Manager	High	High
Waste Collection Staff	Operations User	High	Medium

Residents	Service Users	High	Medium
Developers	System Builders	Medium	High